

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Development of a Dynamic Student Profile Using JavaScript, DOM Manipulation and Event Handling

2. AIM

To develop a dynamic Student Profile webpage using **HTML, CSS and JavaScript**, where student details are taken from input fields, a JavaScript object is created using a class, and the profile is dynamically displayed on the webpage using DOM manipulation.

3. OBJECTIVE

The objectives of this experiment are:

- To create a student profile webpage using HTML.
 - To apply CSS styling to the webpage.
 - To create a JavaScript class named Student.
 - To create objects using a constructor.
 - To accept student details through input fields.
 - To retrieve input values using JavaScript.
 - To manipulate the DOM using JavaScript.
 - To dynamically create HTML elements using createElement().
 - To display dynamic content using textContent.
 - To add elements to the webpage using appendChild().
 - To handle a button click using the onclick event.
 - To dynamically update the student profile.
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4. THEORY

A. JavaScript Class

A **class** is a blueprint used to create objects in JavaScript.

In this experiment, the Student class contains four properties:

- name
- roll
- department
- cgpa

The constructor initializes these properties when a student object is created.

B. DOM Manipulation

DOM stands for **Document Object Model**.

JavaScript can use the DOM to access and modify HTML elements dynamically.

The following methods are used in this experiment:

document.getElementById()

Used to select an HTML element using its id.

Example:

```
document.getElementById("name")
```

.value

Used to retrieve the value entered into an input field.

Example:

```
let name = document.getElementById("name").value;
```

document.createElement()

Creates a new HTML element dynamically.

Example:

```
let heading = document.createElement("h2");
```

.textContent

Adds text content to an HTML element.

Example:

```
heading.textContent = "Student Profile";
```

.appendChild()

Adds a newly created element to another HTML element.

Example:

```
profile.appendChild(heading);
```

C. Event Handling

An event is an action performed by the user, such as clicking a button.

In this program, the onclick event is used:

```
<button onclick="createProfile()">
```

When the **Create Profile** button is clicked, the createProfile() function is executed.

5. ALGORITHM / PROCEDURE

1. Create an HTML document.
2. Add input fields for Name, Roll Number, Department and CGPA.
3. Add a **Create Profile** button.
4. Create a <div> to display the student profile.
5. Add CSS styles to design the webpage.
6. Create a JavaScript class named Student.
7. Define a constructor with name, roll, department and CGPA parameters.
8. Create the createProfile() function.
9. Retrieve values from the input fields using getElementById().
10. Create a Student object using the entered values.
11. Select the profile <div>.

12. Clear any previously displayed profile.
 13. Dynamically create heading and paragraph elements.
 14. Display the student details using `textContent`.
 15. Add the newly created elements to the profile using `appendChild()`.
 16. Click the **Create Profile** button.
 17. Verify that the student profile is displayed dynamically.
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6. PROGRAM

Student_profile.html

```
<!DOCTYPE html>
<html>

<head>

  <title>Student Profile</title>

  <style>

    body {
      font-family: Arial;
      background-color: #f2f2f2;
      text-align: center;
      padding: 30px;
    }

    input {
      padding: 10px;
      margin: 5px;
      width: 250px;
    }

    button {
      padding: 10px 20px;
      background-color: blue;
      color: white;
      border: none;
      cursor: pointer;
```

```
        margin-top: 10px;
    }

    #profile {
        background-color: white;
        width: 350px;
        margin: 25px auto;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0 0 10px gray;
        text-align: left;
    }

    #profile h2 {
        text-align: center;
        color: blue;
    }

    #profile p {
        font-size: 16px;
        margin: 12px;
    }

</style>

</head>

<body>

    <h1>Student Profile</h1>

    <input
        type="text"
        id="name"
        placeholder="Enter Name"
    >
    <br>

    <input
```

```
    type="text"
    id="roll"
    placeholder="Enter Roll Number"
  >
</br>
```

```
<input
  type="text"
  id="dept"
  placeholder="Enter Department"
>
</br>
```

```
<input
  type="text"
  id="cgpa"
  placeholder="Enter CGPA"
>
</br>
```

```
<button onclick="createProfile()">
  Create Profile
</button>
```

```
<div id="profile"></div>
```

```
<script>
```

```
  // Student Class
```

```
  class Student {
```

```
    constructor(name, roll, department, cgpa) {
```

```
      this.name = name;
```

```
      this.roll = roll;
```

```
        this.department = department;

        this.cgpa = cgpa;

    }

}

// Function to create student profile

function createProfile() {

    // Get values from input fields

    let name =
        document.getElementById("name").value;

    let roll =
        document.getElementById("roll").value;

    let dept =
        document.getElementById("dept").value;

    let cgpa =
        document.getElementById("cgpa").value;

    // Create Student object

    let student =
        new Student(name, roll, dept, cgpa);

    // Select profile div

    let profile =
        document.getElementById("profile");
```

```
// Clear previous profile

profile.innerHTML = "";

// Create elements dynamically

let heading =
    document.createElement("h2");

heading.textContent =
    "Student Profile";

let namePara =
    document.createElement("p");

namePara.textContent =
    "Name      : " + student.name;

let rollPara =
    document.createElement("p");

rollPara.textContent =
    "Roll No   : " + student.roll;

let deptPara =
    document.createElement("p");

deptPara.textContent =
    "Department : " + student.department;

let cgpaPara =
    document.createElement("p");
```

```
cgpaPara.textContent =  
    "CGPA      : " + student.cgpa;  
  
    // Add elements to webpage  
  
    profile.appendChild(heading);  
  
    profile.appendChild(namePara);  
  
    profile.appendChild(rollPara);  
  
    profile.appendChild(deptPara);  
  
    profile.appendChild(cgpaPara);  
  
    }  
  
</script>  
  
</body>  
  
</html>
```

7. CODE EXPLANATION

Code	Purpose
<input>	Accepts student information from the user.
id="name"	Identifies the name input field.
id="roll"	Identifies the roll number field.

<code>id="dept"</code>	Identifies the department field.
<code>id="cgpa"</code>	Identifies the CGPA field.
<code><button></code>	Creates the Create Profile button.
<code>onclick</code>	Executes a function when the button is clicked.
<code><div id="profile"></code>	Area where the profile is displayed.
<code>class Student</code>	Creates a Student class.
<code>constructor()</code>	Initializes student properties.
<code>this.name</code>	Stores the student's name.
<code>this.roll</code>	Stores the roll number.
<code>this.department</code>	Stores the department.
<code>this.cgpa</code>	Stores the CGPA.
<code>getElementById()</code>	Selects an HTML element.
<code>.value</code>	Gets the value entered by the user.
<code>new Student()</code>	Creates a Student object.
<code>createElement()</code>	Creates an HTML element dynamically.
<code>textContent</code>	Adds text to an element.
<code>innerHTML = ""</code>	Clears the previous profile.
<code>appendChild()</code>	Adds an element to the webpage.

8. TEST CASES AND EXECUTION DATA

Test Cases	Input / Action	Expected Result	Actual Result	Status
TC-01	Enter student name	Name should be accepted	Accepted	Pass
TC-02	Enter roll number	Roll number should be accepted	Accepted	Pass
TC-03	Enter department	Department should be accepted	Accepted	Pass
TC-04	Enter CGPA	CGPA should be accepted	Accepted	Pass
TC-05	Click Create Profile	Student object should be created	Created	Pass
TC-06	Display profile	Student details should appear	Displayed	Pass
TC-07	Enter new details and click button again	Previous profile should be replaced	Replaced	Pass
TC-08	Check styling	Profile should appear inside styled card	Displayed correctly	Pass

9. OUTPUT

Output :

Student Profile

Create Profile

Student Profile

Name : Ravi Kumar
Roll No : A231265520001
Department : CSE(AI&ML)
CGPA : 8.42

10. RESULT

Thus, a **dynamic Student Profile webpage** was successfully developed using **HTML, CSS and JavaScript**.

The program successfully demonstrated the use of a **JavaScript class, constructor, object creation, DOM manipulation, dynamic element creation, textContent, appendChild() and the onclick event**. The student details entered by the user were successfully displayed dynamically on the webpage
